

Introduction to Tokens and Tokenization

Spoken Tutorial

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<https://spoken-tutorial.org>

Learning Objectives

In this tutorial, we will learn,



System Requirements

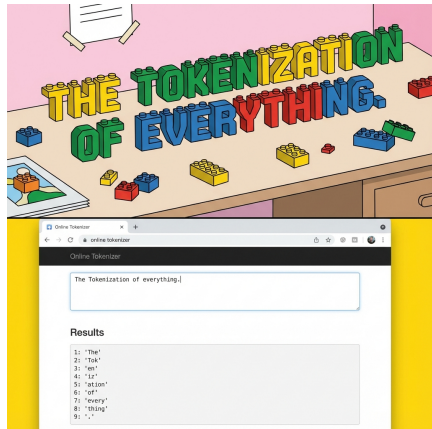


Pre-requisite Slide



What are Tokens?

- ▶ LLMs don't read words like humans.
- ▶ They read tokens.
- ▶ Tokens are sub-word pieces.



How Tokenization Works

- ▶ **Tokenizer is the 'dictionary'.**
- ▶ **Splits raw text into sub-word pieces.**



Why Tokens Matter: Context Windows

- ▶ LLM's 'short-term memory'.
- ▶ Measured in fixed number of tokens.

Hard limit on input context.



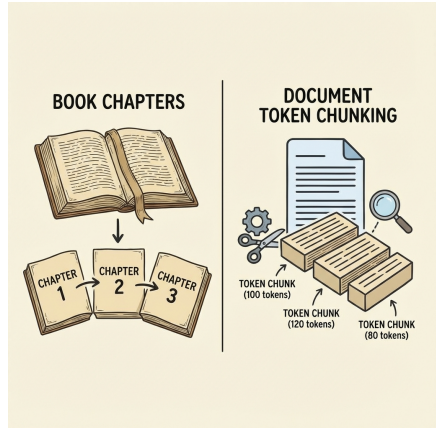
Why Tokens Matter: Cost

- ▶ API providers charge per token.
- ▶ Efficient RAG systems use fewer tokens.
- ▶ Dramatically cheaper to run.



Why Tokens Matter: Chunking

- ▶ **Chunk documents into tokens.**
- ▶ **Ensure chunks fit context window.**



Summary

In this tutorial, we learned that:



Assignment



Domain Expert
(Domain Expert)
Affiliation

Thank you

